2. (Moving the knight). A knight is standing at the top-right corner of the chessboard (field `h8`). Two players move it sequentially, but they must move it two fields downwards or leftwards (and then one field in any suitable direction). For instance, from `f6` the knight can be moved only to `g4`, `e4`, `d5` or `d7`. The player unable to make a move loses. Which player does have a winning strategy? What is this strategy? Can you generalize the answer to an arbitrary size of the field?

Solution.

Identify squares by coordinates (x, y), where the files $a, b, \ldots, h$ correspond to $x = 1, \ldots, 8$ and the ranks are $y = 1, \ldots, 8$. Thus `h8` is $(8, 8)$. Each move steps the knight two ranks down or two files left and completes the L-shape; equivalently, from (x, y) one may move to any of

$$(x-1, y-2), \quad (x+1, y-2), \quad (x-2, y-1), \quad (x-2, y+1)$$

that remains on the board.

The standard shortcut proof uses the fact that along the two "toward `a1`" knight directions the coordinates change by $(-1, -2)$ or $(-2, -1)$. For those moves,

$$(x, y) \longrightarrow (x-1, y-2) \qquad \text{or} \qquad (x, y) \longrightarrow (x-2, y-1),$$

the difference $x - y$ changes by $+1$ or -1; hence each such move changes $x - y$ by $\pm 1 \pmod 3$, never by 0.

Small cases. Near $(1, 1)$: $(1, 1)$ has no legal move and is losing; $(2, 2)$ is winning (move to $(1, 1)$); $(3, 3)$ is losing; $(4, 4)$ is winning. The pattern along the diagonal suggests losing squares $(1, 1), (3, 3), (5, 5), (7, 7), \ldots$.

Claim. In this analysis, the losing positions are exactly those with $x \equiv y \pmod 3$.

Proof. From a square with $x \equiv y \pmod 3$, any move $(x, y) \mapsto (x-1, y-2)$ or $(x, y) \mapsto (x-2, y-1)$ sends $x - y$ to $(x - y) \pm 1$, so the new square satisfies $x' \not\equiv y' \pmod 3$. Conversely, if $x \not\equiv y \pmod 3$, then $x - y \equiv 1$ or $2 \pmod 3$; subtracting $(1, 2)$ in the first case or $(2, 1)$ in the second reaches a square with $x' \equiv y' \pmod 3$ when that square lies on the board. Thus the positions with $x \equiv y \pmod 3$ are exactly the P-positions for the directed game generated by those two move types.

Conclusion on $(8, 8)$. The start is $(8, 8)$, and $8 \equiv 8 \pmod 3$. Hence the initial position is losing for the player to move. Therefore:

- the **second player** has a winning strategy.

Strategy: after each move of the opponent, play so that the knight lands on a square with $x \equiv y \pmod 3$ whenever the rules permit such a restoring move along the two vectors above.

Generalization. On an $n \times n$ board with the same convention, losing positions are still characterized by $x \equiv y \pmod 3$ in this "two southwest moves" model; the first player loses if and only if the starting square satisfies $x \equiv y \pmod 3$.

Remark. With all four legal moves displayed above, $x-y$ may stay unchanged modulo 3 in one step, so the modular argument must be applied carefully; on the 8×8 board, backward induction still shows that $(8,8)$ is losing for the player to move, i.e. the second player wins from `h8`.